CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. Examination (EC)

May 2012

EC310 Optical Communication

Date: 09.05.2012, Wednesday **Time: 01:30 p.m. To 04:30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 (a) What are the advantages of Optical Fibers [04]

(b) When the mean optical power launched into an 8km length of fiber is 120μW, the mean optical power at the fiber output is 3μW. Calculate the signal attenuation per kilometer for the fiber assuming there are no connectors and splices [03]

Q - 2 (a) Briefly discuss different type of losses in optical fiber. [06]

(b) Describe with a neat and labeled sketch the construction of edge emitting LED. [05]

(c) Explain three key transition processes involved in laser action [03]

OR

Q - 2 (a) Explain Fabry Perot resonator cavity for LASER diode [06]

(b) The radiative and nonradiativere combination lifetimes of minority carriers in the active region of a double heterojunction LED are 60ns and 100ns respectively. Determine the total carrier recombination lifetime and power internally generated within the device when the peak emission wavelength is 0.87μm at a drive current of 40mA. [05]

(c) Define dispersion. Discuss material dispersion [03]

Q - 3 (a) Derive the expression for the power launched from the surface emitting LED source to step index fiber for radius of the source greater than and less than the core radius. [06]

(b) With a neat sketch discuss how OTDR is used to take field measurements in optical fibers. [05]

(c) Explain lensing schemes to provide coupling between the source and the diode. [03]

OR

Q - 3 (a) Explain the setup for the eye diagram measurement. [05]

(b) A planar LED is fabricated arsenide which has a from gallium refractive index of 3.6 [05]
(i) Calculate optical power emitted into air as a percentage of the internal optical power for the device when the transmission factor at the crystal-air interface is 0.68.
(ii) When the optical power generated internally is 50% of the electrical power supplied, determine the external power efficiency

(c) Write a short note on Equilibrium Numerical Aperture. [04]

SECTION – II

Q - 4 (a) A point source of light is 12cm below the surface of a large body of water (n=1.33 for water).what is the radius of the largest circle on the water surface through which the light can emerge? [03]

(b) What are skew rays? [02]

(c) List three factors limiting the speed of response of a photodiode. [02]

Q - 5 (a) Explain how Avalanche diode works? Compare APD with PIN diode for sensitivity. What limits the ultimate sensitivity of Avalanche diode? [07]

(b) What are the different types of pre amplifiers? Explain any one of them in detail. [07]

OR

Q - 5 (a) Explain the working of PIN photodiode with quantum efficiency and responsivity equation. [07]

(b) Draw and explain schematic diagram of a typical optical receiver. [07]

Q - 6 (a) Discuss optical power loss model for point to point link with necessary diagram. [05]

(b) A Product sheet for a 2 x 2 single mode biconical tapered coupler with a 40/60 splitting ratio states that the insertion losses are 2.7dB for the 60% channel and 4.7dB for the 40% channel. [05]

(a)If the input power Po = 200μw, find the output levels P_1 and P_2.

(b)Find the excess loss of the coupler.

(c)From the calculated values of P_1 and P_2 , Verify the splitting ratio is 40/60

(c) Draw and explain basic structure of SONET Frame. [04]

OR

Q - 6 (a) What is rise time budget? Explain with necessary equation. [05]

(b) What is the operating principle of mach – zehnder interferometer? Draw the layout of it and explain in brief. [05]

(c) Explain performance of EDFA. [04]